

Matrix transforms in 3D

Basic skills

Q1. a) $\begin{pmatrix} 6 & 3 & 5 \\ -1 & 2 & 3 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 16 \\ 1 \\ -1 \end{pmatrix}$

b) $\begin{pmatrix} 2 & -1 & 0 \\ 1 & 1 & 3 \\ -4 & 3 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ -1 & 0 & 4 \\ -2 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 6 \\ -6 & 3 & 9 \\ -7 & 0 & -8 \end{pmatrix}$

Q2. a) $\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}$ multiply y by 5

b) using formula $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ $\begin{pmatrix} \cos 150^\circ & -\sin 150^\circ \\ \sin 150^\circ & \cos 150^\circ \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$

c) $y = \sqrt{3}x$ so for formula $\tan \theta = \sqrt{3}$ $\theta = 60^\circ$
 $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$ so $\begin{pmatrix} \cos 120^\circ & \sin 120^\circ \\ \sin 120^\circ & -\cos 120^\circ \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$

Q3. Reflection in $x=0$ changes sign of x -coordinate, so $A' = (-5, -2, 3)$

Q4. a) $\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\times$ sign change
Reflection in $x=0$

b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ stretch in y of 5

c) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = 3I_3$ enlargement SF: 3

d) $\begin{pmatrix} \cos \frac{\pi}{4} & 0 & \sin \frac{\pi}{4} \\ 0 & 1 & 0 \\ -\sin \frac{\pi}{4} & 0 & \cos \frac{\pi}{4} \end{pmatrix}$ using formula
 $\begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$
 rotation of $\frac{\pi}{4}$ AC
 around y -axis

e) $\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$ using formula
 $\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\cos \theta = \frac{\sqrt{3}}{2}$, $\sin \theta = \frac{1}{2}$

so $\theta = 30^\circ$ so 30° AC rotation around z -axis

Q5. $\Theta = 180^\circ$ using formula

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \Theta & -\sin \Theta \\ 0 & \sin \Theta & \cos \Theta \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos 180^\circ & -\sin 180^\circ \\ 0 & \sin 180^\circ & \cos 180^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Q6. A coordinate cannot be reflected in a line as a set (of circular points) are perpendicular we would have to choose a point on a straight line. would need vector calculations for perpendicularity

Q7. $A' = (4, -3, 1)$ as $\frac{1}{1} \frac{1}{2} \frac{1}{3} =$

Q8. As it affects the origin, any $M \underline{0} = \underline{0}$ however here would need $M \underline{0} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$, not possible

Translation techniques needed

Q9. a) As each change sign

Guess $M = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = -I_3$

So $Mx = -I_3 x = -x$ as need to change signs of coordinates

b) $X = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ $Y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $Z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

$$M = XYZ = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

so Yes true for this

c) For this technique of any order each step can only affect one coordinate

Should do $M = \underset{\substack{\uparrow \\ \text{order}}}{Z} Y X$